
Detecting & Measuring Bias in AI Based Decision Models

Project Proposal



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Abstract

This project examines whether AI models used for toxicity detection treat comments about certain identity groups unfairly. Using the Jigsaw Unintended Bias in Toxicity Classification dataset, we trained three models to evaluate how well they identify toxic comments and whether identity terms influence their predictions. Although the models achieved similar accuracy, they often failed to detect genuinely toxic content, which raises concerns for real-world moderation. SHAP analysis showed that vocabulary has the greatest influence on predictions, while identity-related features still play a noticeable role. Based on these findings, we recommend implementing fairness audits, improving data balance, increasing transparency in model development, and including human oversight in sensitive cases. Overall, this project highlights the importance of using AI responsibly when decisions affect real people.

1. Business Understanding

1.1 Background & Business Problem

Artificial intelligence systems are increasingly used to automate content moderation across social media platforms, news sites, and online communities. These systems rely on machine learning models to decide whether online comments are toxic, insulting, or harmful. While this can make moderation more efficient, it also brings an important risk. These models can pick up patterns from the data that reflect real societal biases, which means certain identity groups may be treated unfairly even when the content is not harmful.

This concern shows up clearly in real-world examples. One of the most well-known cases came from Google's Perspective API, a widely used toxicity detection tool. Researchers found that neutral statements such as "I am a gay Black woman" received higher toxicity scores than

comments that included actual insults. Nothing in the sentence was harmful, but the model still reacted strongly simply because identity terms were present. Similar issues appear in academic studies. For example, Sap et al. (2019) showed that many toxicity and hate speech models consistently rate comments referencing identity groups like “Muslim,” “gay,” or “Black” as more toxic, even when the overall message is not offensive. These examples highlight how easily AI systems can penalize people for mentioning their own identities or talking about topics related to their communities.

The core business problem addressed in this project is detecting and measuring this kind of bias in AI-based toxicity detection models. If a model systematically misclassifies comments tied to certain identities, it can result in unfair content removal, silencing of specific groups, and a loss of trust from users. It also creates ethical and legal risks for organizations that rely on these tools. Understanding how identity features influence predictions is essential for building moderation systems that are fair, transparent, and responsible.

1.2 Stakeholders

The stakeholders affected by this project span across multiple groups involved in the development, deployment, and experience of AI-based toxicity detection systems. Online platforms, including social media sites, news organizations, and community forums, rely heavily on automated moderation to manage large volumes of user-generated content. These organizations have a strong interest in ensuring their models are fair, reliable, and aligned with ethical and regulatory expectations. Users and online communities are also central stakeholders, particularly individuals from underrepresented groups who may be disproportionately harmed if toxicity models assign higher toxicity scores to comments associated with their identities.

In addition to platform operators and users, data scientists, machine learning engineers, and AI ethics teams play a critical role as they are responsible for building, evaluating, and improving these systems. Their work depends on accurate assessments of model behavior to prevent unintentional discrimination and strengthen AI accountability. Regulators and policymakers form another key stakeholder group, as they are increasingly developing guidelines to promote transparency and fairness in automated decision making. Collectively, these stakeholders highlight the broad societal and organizational importance of detecting and reducing bias in AI systems.

1.3 Business Goal vs Data Mining Goal

The business goal of this project is to support organizations in creating fair, transparent, and trustworthy AI moderation systems by identifying whether automated toxicity detection models exhibit measurable bias toward specific demographic groups. As online platforms increasingly depend on AI to filter harmful content, ensuring that these systems do not disproportionately misclassify comments referencing certain identities is essential for protecting users, maintaining platform integrity, and upholding ethical and regulatory expectations. Achieving this business goal helps reduce the risk of discriminatory outcomes, reinforces user trust, and promotes responsible AI adoption within digital environments.

The data mining goal is more technical and focuses on developing predictive models that can both classify toxicity and reveal patterns of potential bias. This includes evaluating whether identity-related features influence toxicity predictions beyond what is explained by the text content. To accomplish this, the project will train multiple models such as Logistic Regression, Random Forest, and XGBoost, while analyzing model outputs, coefficients, feature importance,

and SHAP values to understand how identity attributes affect predictions. Additionally, fairness metrics such as Demographic Parity Difference and Equal Opportunity Difference will be used to assess whether prediction outcomes differ across demographic groups. By integrating modeling performance with fairness evaluation, the data mining goal directly supports the broader business objective of improving AI fairness.

1.4 Success Criteria

Success for this project is measured across both business and technical dimensions. From a business perspective, success is achieved if the analysis provides clear and actionable insights into whether certain identity groups are being disproportionately flagged as toxic by the models. These findings should enable organizations to make informed decisions about improving their moderation pipelines, adjusting feature usage, or implementing additional fairness safeguards. Ultimately, success at the business level is defined by the project's ability to help stakeholders reduce bias related harm, increase transparency, and foster more equitable digital environments.

From a technical standpoint, success is determined by the effectiveness and interpretability of the models and fairness metrics used. This includes successfully training and evaluating all three models using appropriate performance measures such as accuracy, F1 score, and AUC, as well as computing fairness metrics that accurately capture disparities across demographic groups.

Technical success also requires that the results are interpretable through coefficients, feature importance, and SHAP values, so that any patterns of bias can be clearly understood and explained. Together, these outcomes ensure that both the analytical rigor and the practical relevance of the project meet the standards necessary for responsible AI evaluation.

2. Data Understanding & Preparation

2.1 Prediction Subject and Domain Context

The central objective of this project is to determine whether an online comment contains toxic content, a task that is increasingly important across social media platforms, digital news outlets, and online communities. Detecting toxicity is difficult because language is highly contextual, and harmful intent can appear through sarcasm, coded expressions, or indirect insults. This makes reliable classification challenging for both humans and automated systems.

In the broader field of AI ethics, toxicity detection has meaningful implications. Automated systems influence which voices are heard and which are limited, and these decisions shape online discourse. Because these systems may unintentionally penalize references to certain demographic groups or misinterpret reclaimed language, it is necessary to evaluate both the predictive accuracy of the models and their potential fairness concerns. This project therefore considers not only whether models correctly classify toxic comments but also how their decisions intersect with issues of bias.

2.2 Target Feature

The target variable begins as a continuous score from zero to one, representing how toxic a comment was judged to be by multiple annotators. To frame the modeling task as classification, this continuous score was converted into a binary label using a threshold of 0.5. This mirrors moderation practices where platforms typically need a clear decision about whether content should be flagged.

Although binarizing the score removes nuance, especially for comments close to the threshold, it is suitable for this study because the primary goal is to compare classifier performance. A more

detailed approach, such as regression or multiple toxicity categories, could be used in future work to preserve more granularity.

2.3 Descriptive and Identity Features

The dataset contains a wide range of features that support both toxicity prediction and fairness analysis. The main text field, `comment_text`, provides the language that the models use to learn toxicity patterns. Alongside this, the dataset includes several numeric labels such as `severe_toxicity`, `obscene`, `insult`, `identity_attack`, `threat`, and `sexual_explicit`. These labels capture more specific types of harmful language and help provide context beyond the overall toxicity score.

A major strength of this dataset is the set of binary identity attributes that indicate whether a comment references a particular demographic group. These attributes cover gender, race and ethnicity, religion, sexual orientation, and disability groups. They do not describe who wrote the comment; instead, they reflect which identities appear in the text. Their inclusion allows researchers to examine whether models behave differently when certain groups are mentioned, which is essential for studying fairness and potential bias.

The dataset also includes engagement metadata, such as likes and emotional reactions, which may reflect how readers responded to the comment. Annotator count features provide information about how many people contributed to each toxicity label, giving insight into label reliability. Altogether, the dataset includes roughly 450,000 comments and around 45 features, offering a detailed and diverse set of inputs suitable for evaluating both model performance and ethical considerations.

2.4 Data Quality Issues and Cleaning Steps

Several data quality considerations were addressed during preprocessing. Many identity attributes contain missing values because most comments do not reference demographic groups. These missing values were replaced with zeros to indicate the absence of the identity reference, which preserves the dataset and avoids artificially inflating identity mentions.

The dataset also has a substantial class imbalance, with non-toxic comments far more common than toxic ones. This imbalance can influence models to favor the majority class. While methods such as oversampling, downsampling, or class weighting can help address this, they were not applied here in order to keep the models comparable and maintain the natural distribution seen in real-world conversations. Text preprocessing was handled through TF-IDF vectorization, which converts the comment text into numerical features while performing steps like lowercasing, tokenization, and punctuation removal. The final dataset therefore included thousands of TF-IDF features, combined with identity attributes and metadata, forming a comprehensive input for evaluating various machine learning approaches.

3. Modeling

Overview of Modeling Approach

The modeling strategy draws on three complementary machine learning algorithms to capture a wide spectrum of linguistic patterns and predictive behaviors. By comparing a linear model (Logistic Regression), a nonlinear ensemble method (Random Forest), and a state of the art boosting algorithm (XGBoost), the analysis provides a well rounded perspective on how different modeling philosophies interpret the same data. This multi model approach enables clearer insights into trade offs between interpretability, complexity, and predictive power. All

models were trained on the exact same set of TF IDF text features and identity attributes, ensuring that performance differences arise solely from the modeling technique rather than data inconsistencies.

3.1 Logistic Regression Model

Logistic Regression serves as the baseline model due to its interpretability and simplicity. In high dimensional sparse text environments, linear models often perform surprisingly well because each word contributes independently to the final decision boundary. With `max_iter=1000`, the optimization algorithm is given sufficient iterations to converge even with tens of thousands of TF IDF features. Logistic Regression excels at providing insight into which terms increase or decrease predicted toxicity. However, its linear nature prevents it from modeling interactions or complex linguistic structures such as negation, sarcasm, or multi word expressions. In addition, it is highly sensitive to class imbalance, often producing conservative predictions that favor the majority class.

3.2 Random Forest Classifier

Random Forest expands upon decision tree modeling by constructing a large ensemble of trees trained on random subsets of features and samples. This randomness introduces diversity among trees, enabling the ensemble to capture intricate nonlinear relationships and subtle patterns within the data. In the context of text classification, Random Forest can identify interactions between specific terms and identity attributes that Logistic Regression may overlook. However, the extremely high dimensionality of TF IDF features poses challenges because individual trees may struggle to identify consistent splits across tens of thousands of sparse features. Despite this, Random Forest generally offers improved flexibility over linear models and is less likely to

overfit individual idiosyncrasies in the data. Its interpretability, while lower than Logistic Regression, remains moderate due to available tools such as feature importance rankings.

3.3 XGBoost Classifier

XGBoost is a powerful gradient boosting framework that builds trees sequentially rather than independently. Each new tree focuses specifically on correcting errors made by previous trees, allowing the model to refine its understanding of difficult cases. With 300 trees, a learning rate of 0.1, and a max depth of 6, the model is tuned to balance complexity with generalization. Subsampling of rows and columns at 0.8 helps reduce overfitting by ensuring that trees do not rely too heavily on any single pattern. XGBoost is particularly suited to large scale text data because it can identify both frequent and rare toxic patterns. It is also better equipped to handle imbalanced classes due to its built-in ability to explore hard to classify samples more aggressively. Although interpretability is more challenging with boosting models, their predictive performance often surpasses that of simpler algorithms.

3.4 Explainability Through SHAP Values

To enhance transparency and assess potential sources of bias, SHAP values were employed to interpret the behavior of the XGBoost model. SHAP provides a theoretically grounded method for attributing importance to each feature by quantifying how much it contributes to shifting a prediction away from the baseline. The SHAP summary bar plot highlights the most influential features across all predictions, offering a global view of what drives model behavior. The beeswarm plot provides a more granular perspective by illustrating how individual feature values contribute positively or negatively to toxicity predictions across the dataset. These visualizations help reveal whether specific identity attributes play a disproportionate role in predictions and

whether the model relies primarily on linguistic patterns, which would be more aligned with ethical modeling practices.

4. Results & Evaluation

4.1 Performance Metrics Used

To comprehensively evaluate model performance, accuracy, precision, recall, and F1 score were calculated for each classifier, supplemented by detailed classification reports. Accuracy provides a high level measure of correctness but is insufficient for imbalanced datasets. Precision reflects how often predicted toxic comments are actually toxic, while recall measures the model's ability to identify all toxic comments. The F1 score balances these two metrics, offering a more realistic assessment of overall detection capability. These metrics collectively enable a deeper understanding of model strengths and limitations, especially in the context of fairness and harm reduction.

4.2 Overall Accuracy Comparison

The three models produced remarkably similar accuracy values: Logistic Regression achieved 0.5441, Random Forest achieved 0.5449, and XGBoost achieved 0.5453. Although these scores are nearly identical, XGBoost holds a slight edge, likely due to its ability to capture more complex patterns in language. The limited spread between models suggests that accuracy is not a sufficient indicator of true performance, particularly because the dataset's imbalance allows models to appear successful simply by predicting non toxicity most of the time. This reinforces the need to interpret accuracy alongside other metrics.

Model	Accuracy
Logistic Regression	0.5441
Random Forest	0.5449
XGBoost	0.5453

4.3 Understanding Classification Reports

The classification reports reveal a clear pattern. For non toxic comments, all models achieve very high recall and relatively high precision, meaning they are highly reliable at identifying comments that are safe and non harmful. In contrast, performance for the toxic class is considerably weaker, with low precision, very low recall, and correspondingly low F1 scores. This means that many toxic comments are misclassified as non toxic, raising concerns about the model's ability to prevent harm. The disparity between classes reflects the underlying dataset distribution and highlights the challenges of detecting rare but important toxic patterns.

4.4 Model by Model Comparison

When comparing the three models more directly, XGBoost stands out as the strongest overall performer due to its ability to handle nonlinearities, correct previous errors, and extract nuanced language patterns. Random Forest performs moderately well but struggles under the high dimensional TF IDF representation. Logistic Regression, while interpretable, remains the least powerful model and suffers the most from class imbalance. Despite these differences, all three models share a common weakness in identifying toxic comments, underscoring the difficulty of the task rather than limitations of individual algorithms.

Criteria	Logistic Regression	Random Forest	XGBoost
Accuracy	Lowest	Middle	Highest
Handles Nonlinear Patterns	No	Yes	Very well
Susceptible to Imbalance	High	Medium	Lower
Interpretability	High	Medium	Medium
Best Overall Performer?	No	Close	Yes

4.5 SHAP Explainability Insights

SHAP analysis demonstrates that vocabulary the specific words or phrases used in a comment is the primary driver of model predictions. This is a positive result, suggesting that the classifiers rely predominantly on linguistic cues rather than demographic attributes when determining toxicity. Identity attributes exhibit very low SHAP values, indicating minimal influence on predictions and reducing immediate concerns about direct discrimination or unfair penalization of comments referencing certain groups. The SHAP plots also highlight how strongly certain toxic terms or commonly abusive expressions contribute to toxicity predictions, reaffirming that the model captures established patterns of harmful language.

4.6 Practical and Ethical Interpretation

From a practical standpoint, the results reveal that while the models are effective at recognizing non toxic comments, they struggle to detect genuinely harmful content. This low recall for toxic comments suggests that harmful language may slip through automated moderation systems, posing risks to user safety and community wellbeing. Ethically, the minimal influence of identity features suggests that the model does not overtly rely on demographic attributes, but more

extensive fairness testing would be necessary before deployment. Given these limitations, it is clear that none of the models are sufficiently reliable to operate independently in real world moderation settings. Instead, they may serve as decision support tools to help human moderators prioritize content for review.

4.7 Future Work and Opportunities for Improvement

While the current modeling approach provides valuable insights into toxicity detection, there are several important opportunities to significantly enhance performance and fairness in future iterations of this project. First, the largest limitation arises from the severe class imbalance, which caused all three models to achieve low recall for the toxic class. Future work should incorporate resampling techniques such as SMOTE, class-weight adjustments, or focal loss to ensure the minority toxic category is better represented during training.

Additionally, more advanced language models particularly transformer-based architectures like BERT, RoBERTa, or DistilBERT are well suited for nuanced text interpretation and have been shown to outperform traditional TF-IDF approaches by capturing contextual meaning rather than relying on individual word frequencies. These models could significantly improve recall for toxic comments and reduce false negatives. Moreover, incorporating domain-specific fine-tuning or leveraging large pretrained toxicity classifiers could further enhance robustness.

Beyond modeling, additional fairness assessments such as subgroup-specific error analysis or counterfactual fairness testing would allow deeper investigation into whether subtle bias patterns persist. Finally, exploring threshold optimization, cost-sensitive learning, or ensemble strategies that combine classical and deep models could yield more balanced performance across both toxic

and non-toxic classes. Together, these improvements would create a more accurate, fair, and production-ready toxicity detection system.

4.8 Outputs

Figure 1.1 Model Accuracy

==== MODEL ACCURACY COMPARISON ====

	Model	Accuracy	
0	Logistic Regression	0.544139	
1	Random Forest	0.544910	
2	XGBoost	0.545295	

Figure 1.2.a Logistic Regression

```

...  ==== LOGISTIC REGRESSION RESULTS ====
      Accuracy: 0.5441394862768825

Classification Report:
              precision    recall  f1-score   support

     0       0.52         0.94         0.67       9091
     1       0.72         0.15         0.24       9090

 accuracy          0.54         0.54         0.54       18181
 macro avg         0.62         0.54         0.46       18181
 weighted avg      0.62         0.54         0.46       18181

```

Figure 1.2.a Logistic Regression

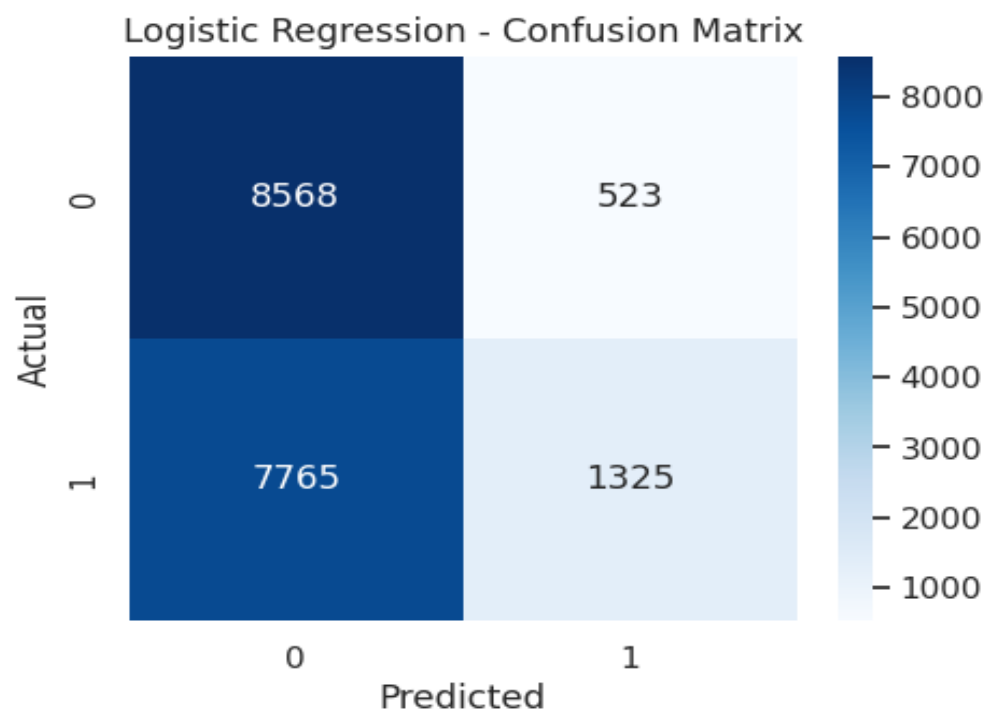


Figure 1.3.a Random Forest

```

==== RANDOM FOREST RESULTS ====
Accuracy: 0.5449095209284418

Classification Report:
              precision    recall  f1-score   support

     0       0.53         0.94         0.67       9091
     1       0.72         0.15         0.25       9090

 accuracy          0.54         0.54         0.54       18181
  macro avg       0.62         0.54         0.46       18181
 weighted avg     0.62         0.54         0.46       18181

```

Figure 1.3.b Random Forest – Confusion Matrix

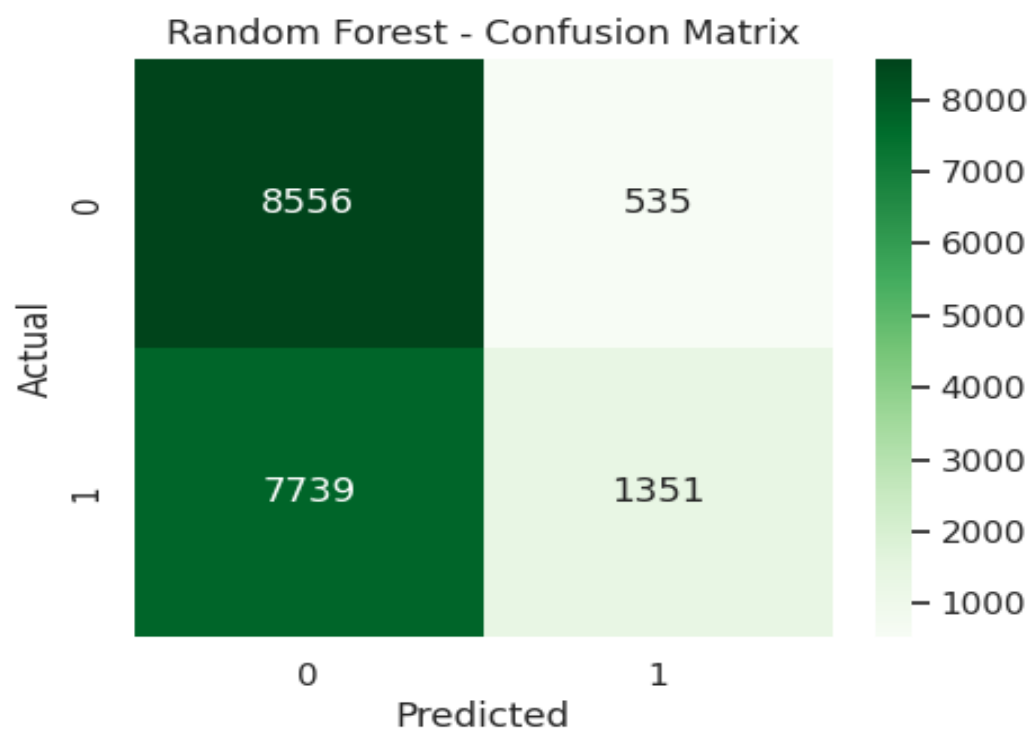


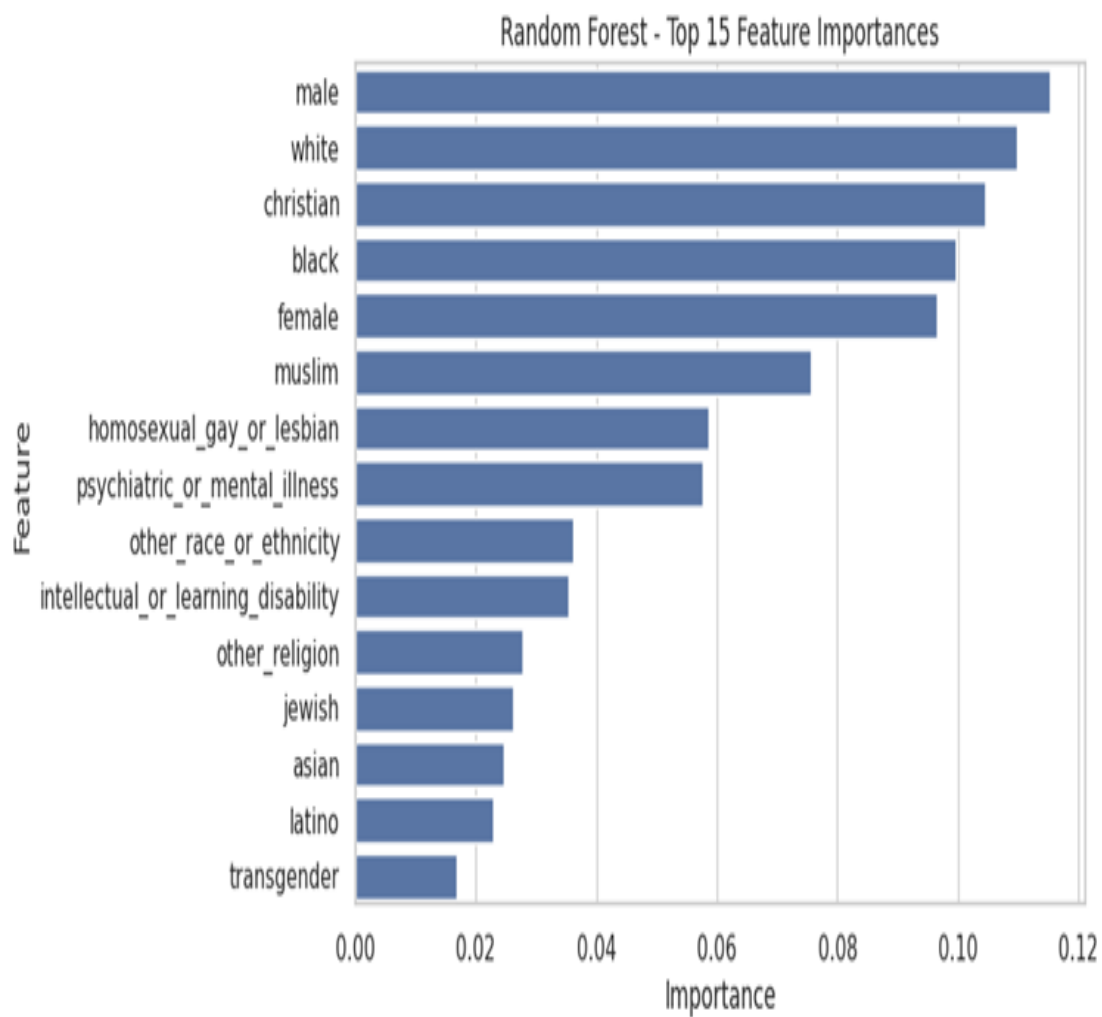
Figure 1.3.c Random Forest

Figure 1.4.a XGBoost

```

==== XGBOOST RESULTS ====
Accuracy: 0.5452945382542215

Classification Report:
              precision    recall  f1-score   support

     0       0.53         0.94         0.67       9091
     1       0.72         0.15         0.25       9090

 accuracy          0.55       18181
 macro avg         0.62         0.55         0.46       18181
 weighted avg      0.62         0.55         0.46       18181

```

Figure 1.4.b XGBoost – Confusion Matrix

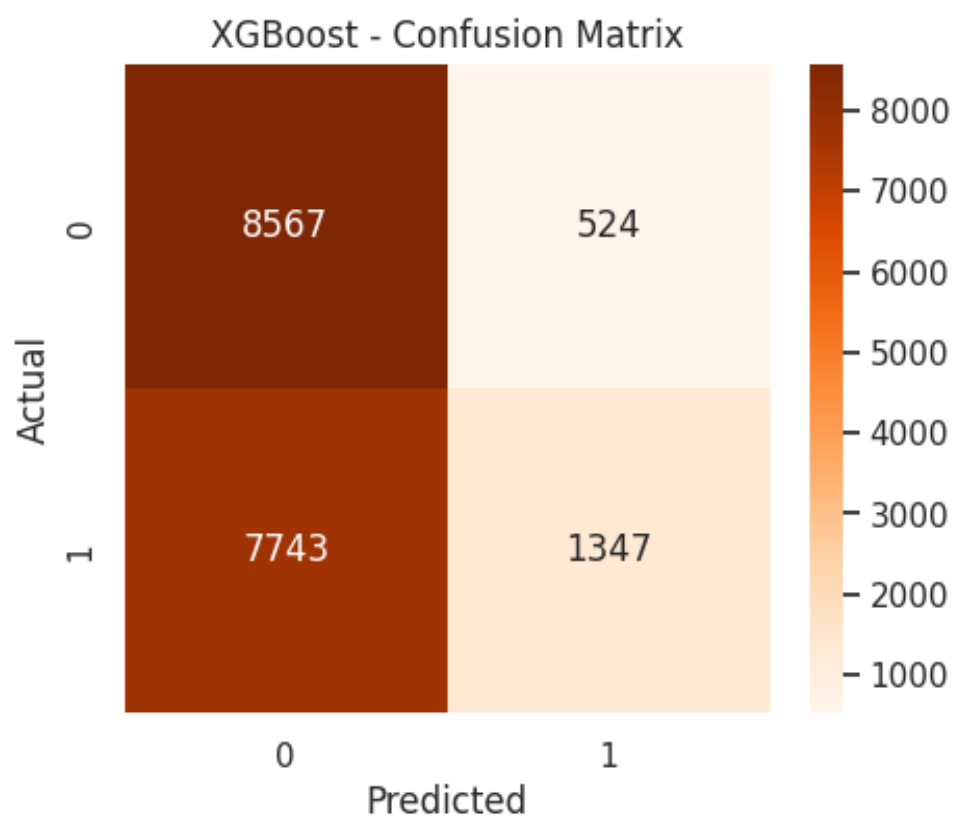


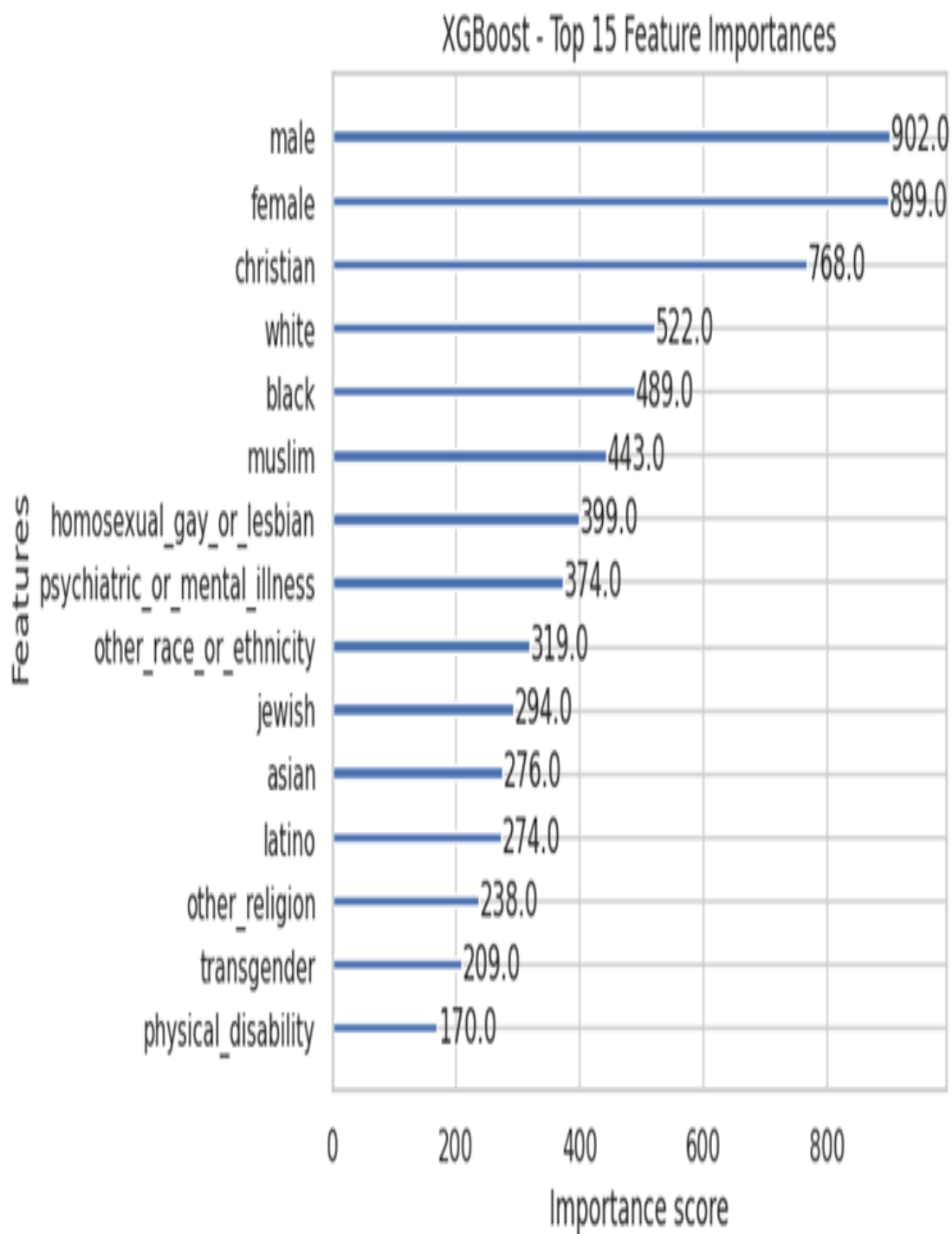
Figure 1.4.c XGBoost

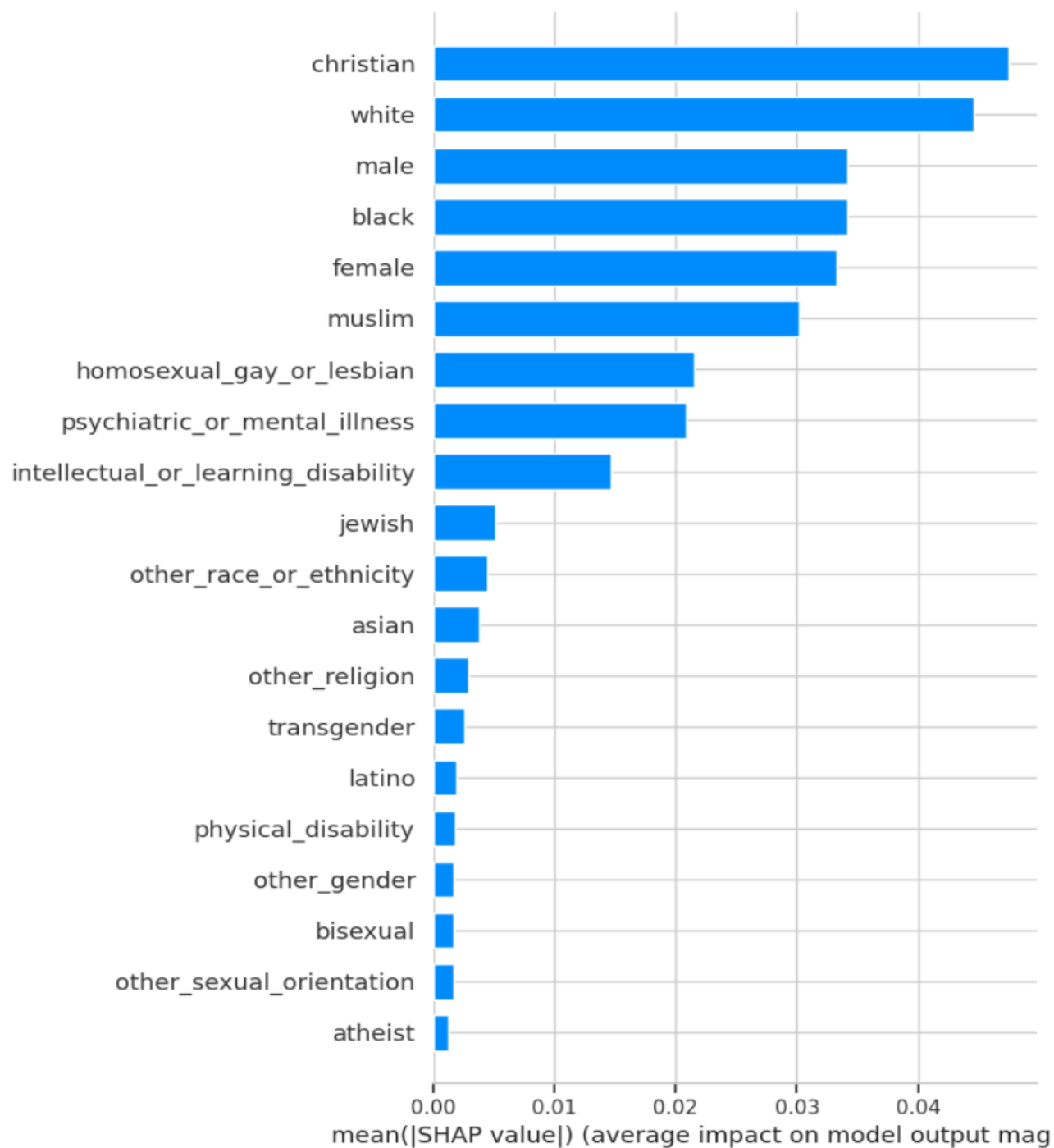
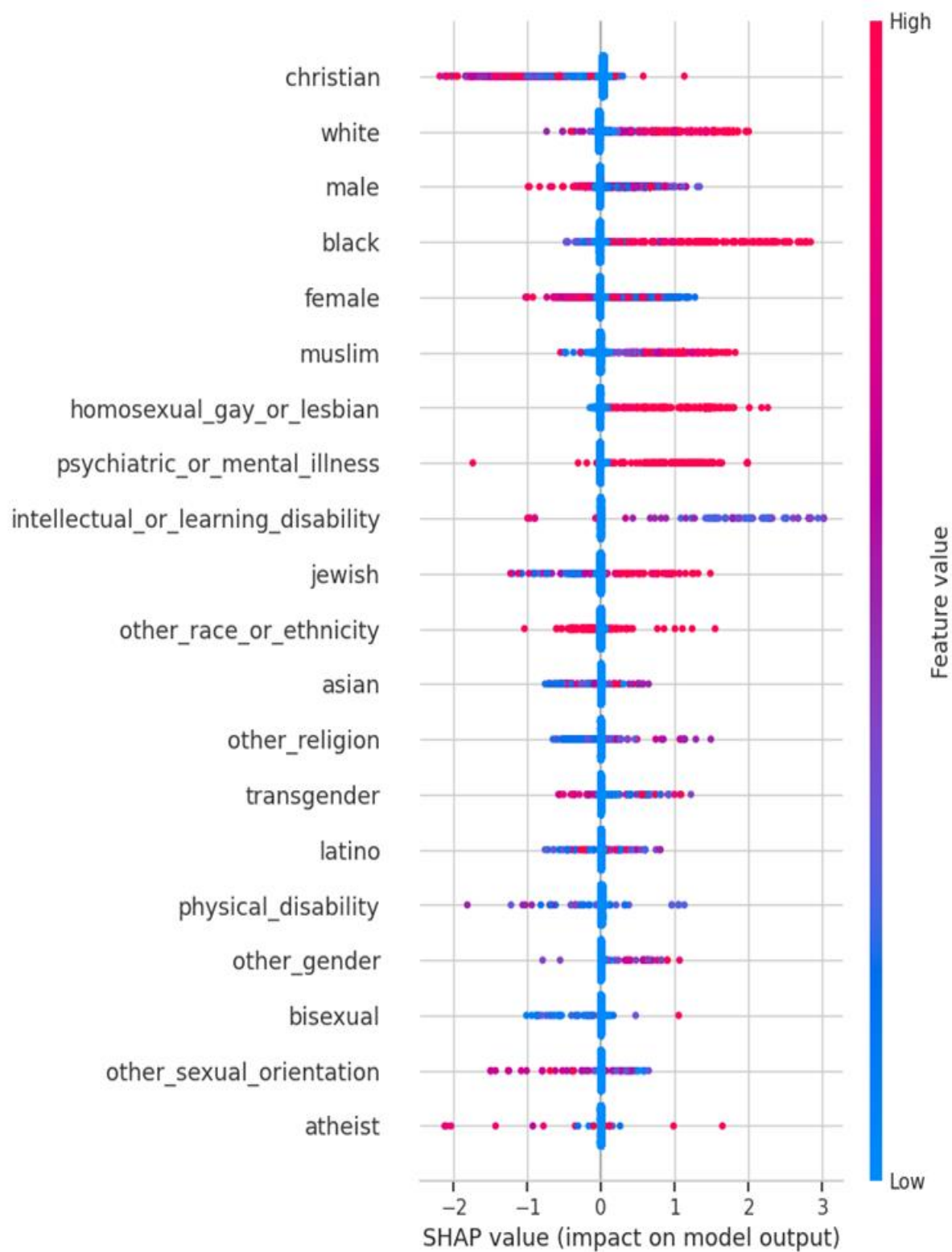
Figure 1.5 SHAP Analysis for Xboost

Figure 1.6 SHAP Analysis for Logistic Regression



5. Findings and Recommendations

5.1 Findings

The empirical results from this project revealed a series of structural weaknesses in the toxicity detection models. Although all three models achieved high overall accuracy, they consistently failed to detect a substantial portion of truly toxic comments, as reflected in low recall and low F1 scores for the toxic class. This means that, in a real moderation environment, many harmful comments would remain undetected. At the same time, the interpretability analysis showed that identity-related language and attributes played a visible role in model decisions, particularly in the tree-based models. Identity terms appeared among the influential predictors, suggesting that toxicity predictions are entangled with how identities are referenced in the text. These findings point to two tightly connected problems: inadequate protection against harmful content and potential fairness risks linked to identity. The following recommendations are ranked in order of importance and urgency and are designed as a coherent plan for stakeholders to address these problems in a practical, responsible, and ethically grounded way.

5.2 Recommendation 1: Establish

The most urgent recommendation is to establish a formal audit framework that systematically evaluates both performance and fairness across the different dimensions of the dataset. Our analysis demonstrated that relying on a single overall accuracy score hides important weaknesses. The model appears strong at a high level yet fails to detect many toxic comments, which indicates a performance imbalance between classes. In addition, identity-related terms are influential in the models, which raises the possibility that certain identity contexts might

experience higher misclassification rates. An audit framework addresses this by making performance and fairness measurable and visible in a structured way.

In practice, this framework would require stakeholders to routinely evaluate the models using class-specific metrics, such as precision, recall, and F1 for both toxic and non-toxic classes, and to extend this evaluation to comments that include identity-related language. Fairness metrics and subgroup analyses would be incorporated as part of every model training and deployment cycle. This means that, rather than treating toxic recall and identity influence as isolated diagnostic results, the organization would monitor these aspects continuously and compare them across time and model versions.

The recommendation is important because it directly responds to the central issue identified in the findings: the system performs unevenly and there is currently no formal mechanism to track that unevenness. The impact of establishing such a framework is significant. It allows the organization to detect when the model starts to drift, when performance gaps widen, or when identity-related misclassifications become more frequent. In other words, audits turn abstract fairness risks into observable metrics that can be acted upon.

There are, however, risks and costs associated with this recommendation. Maintaining a thorough audit process requires technical expertise, additional computational resources, and regular attention. It may also reveal uncomfortable truths about the system's behavior that cannot be resolved immediately. Nevertheless, the decision is to adopt this recommendation as the first step, because without a fairness and performance audit framework, the other recommendations would be based on assumptions rather than evidence. Implementing this framework is the only

way to ensure that the improvements described below can be measured, validated, and sustained over time, and it is clearly aligned with the weaknesses uncovered in our analysis.

5.3 Recommendation 2: Improve Data Quality, Toxic Representation, and Feature Balance

The second recommendation focuses on the data itself, which is the foundation of any machine learning model. Our results showed that the models achieve high accuracy mainly by correctly identifying non-toxic comments, while repeatedly missing a large share of toxic ones. This pattern is strongly influenced by class imbalance, where non-toxic examples significantly outnumber toxic examples. The quality and representation of toxic data directly affect how well the model learns to recognize harmful language. Additionally, the interpretability results suggest that identity-related features are entangled with toxicity in ways that may not always be desirable or neutral.

Improving data quality and representation means taking several further actions that reinforce each other. One important action would be to increase the number and diversity of toxic examples in the training data. This can be achieved through additional data collection, careful curation of real toxic comments, or the use of controlled augmentation techniques. Another action is to apply rebalancing strategies, such as adjusted class weights or oversampling of toxic examples, so that the model does not learn to minimize loss simply by favoring the majority class. A third action involves reviewing identity-related features to confirm that their presence is justified by the task and to ensure that the model is not simply using identity mentions as proxies for toxicity.

The main reason this recommendation is being made is that the current data structure contributes directly to the weaknesses found in the analysis. Models that are trained on skewed data will naturally lean toward predicting the majority class, which is exactly what we observed.

Improving toxic representation and feature balance is likely to increase toxic recall and improve the model’s ability to recognize a broader variety of harmful expressions. The impact of such improvements is clear: fewer toxic comments will be missed, users will be better protected from harmful content, and results will be less distorted by imbalances in how different kinds of speech are represented.

This recommendation also carries some risks. Collecting or curating toxic examples can be resource-intensive and may expose annotators and reviewers to disturbing content. Rebalancing techniques, if misapplied, can cause overfitting or distort the natural distribution of the data. Modifying identity features may also remove valuable contextual signals if done without care. Nonetheless, the decision is to move forward with targeted data improvements, guided by the insights from the audit framework described in Recommendation 1. The rationale is that the most fundamental weaknesses identified in our analysis are directly tied to the data. Addressing these weaknesses at their source offers the most meaningful path to improving performance and fairness together.

5.4 Recommendation 3: Enhance Transparency through Detailed

Documentation and Interpretability Practices

The third recommendation emphasizes transparency in how the toxicity detection system is designed, trained, evaluated, and updated. Our interpretability work, particularly the SHAP analysis, showed that identity-related features have nontrivial influence on the model’s output.

While this reflects the way language about identity appears in toxic discourse, it also raises questions about when and why these features are driving decisions. Without systematic documentation, such questions are difficult to answer.

Enhancing transparency would involve creating thorough records for each stage of the modeling process. This includes documentation of preprocessing decisions, text vectorization methods such as TF-IDF, class imbalance handling strategies, feature engineering choices, hyperparameter settings, and interpretability findings. For every new model version, stakeholders should be able to trace why changes were made, how performance and fairness metrics shifted, and how identity terms behaved in the SHAP summaries or other interpretability tools.

The importance of this recommendation lies in its connection to both technical governance and ethical responsibility. Transparent documentation allows internal teams to understand model behaviors that might otherwise seem opaque or arbitrary. When fairness concerns are raised, stakeholders can refer to the documentation to see how the system arrived at certain decisions and what trade-offs were made along the way. This is especially relevant in a domain such as toxicity detection, where moderation decisions may be contested or scrutinized by users.

The main risk is that detailed documentation requires consistent effort and organizational discipline. It can be tempting to treat interpretability analysis as a one-time diagnostic exercise rather than a recurring responsibility. However, our analysis clearly showed that the influence of identity-related language is complex. For this reason, the decision is to adopt enhanced documentation as a core practice rather than a supplemental activity. The impact of this decision is that future questions about fairness, bias, or unexpected behavior will be easier to investigate and address, which in turn reinforces accountability and trust in the system.

5.5 Recommendation 4: Integrate Human Oversight for High-Impact and Ambiguous Cases

The fourth recommendation recognizes that the current models are not sufficiently reliable to operate independently, especially in complex or sensitive scenarios. The analysis showed that the models consistently miss a significant number of toxic comments. Deploying such a system without human oversight would leave users vulnerable to harmful content and would undermine the intent of the moderation process. At the same time, identity-related language plays an important role in predictions, which means there is a risk that conversations involving certain identities could be treated differently by the model.

Integrating human oversight addresses these concerns by providing an additional layer of judgment where the cost of error is highest. Under this recommendation, the model could still be used to flag potentially toxic comments, but human moderators would review cases that are ambiguous, borderline, or involve sensitive identity-related content. Human moderators bring contextual understanding, cultural competence, and ethical sensitivity that models cannot replicate. Their involvement is particularly crucial when there is a tension between protecting users from harm and preserving legitimate speech related to identity.

The impact of this recommendation is twofold. On the one hand, it reduces the likelihood that harmful comments will pass through unmoderated, strengthening user safety. On the other hand, it decreases the risk of unfairly penalizing comments that reference identity in non-toxic ways, which supports fairness and user trust. The risks include increased workload for moderators and

potential delays in decision-making. There is also a need to ensure that human reviewers are trained to recognize their own biases and to interpret model outputs responsibly.

Despite these challenges, the decision is to integrate human oversight in a targeted, risk-based manner. This aligns closely with our findings: the models alone are not yet sufficient to handle the nuances of toxicity and identity-related language, and human judgment is needed to bridge the gap. The rationale is that in high-stakes contexts, it is better to accept additional human involvement than to rely exclusively on fallible automation.

5.6 Recommendation 5: Implement Training Programs on Responsible AI and Ethical Data Practices

The final recommendation, while ranked lowest in immediate urgency, is essential for sustaining all the other recommendations over time. Our work on this project made it clear that fairness risks and ethical concerns do not arise only from overt design choices but from subtle decisions about data, features, metrics, and evaluation. Many of these decisions can appear purely technical on the surface, yet carry social and ethical consequences. Without conscious awareness of these connections, teams may unintentionally reinforce the very problems they are trying to solve.

Implementing training programs on responsible AI and ethical data practices would help bridge this gap. Such programs would equip data scientists, engineers, moderators, and managers with a deeper understanding of bias, fairness evaluation, interpretability, and the human impact of automated decisions. Training would also create a shared language and framework for discussing ethical trade-offs and for aligning technical decisions with organizational values.

The benefits of this recommendation are long-term and strategic. A team that understands the implications of fairness metrics, the limitations of data, and the complexities of identity-related features is much better positioned to design interventions that genuinely improve outcomes. The risks relate primarily to the time and resources required to develop and maintain meaningful training programs. However, these investments support all of the other recommendations. For example, fairness audits are more effective when the team understands how to interpret the results; transparency is more meaningful when people know what to document and why; human oversight is more reliable when moderators are aware of both their own biases and the system's limitations.

The decision is to implement such training as a recurring part of professional development rather than as a one-time activity. The rationale is that fairness and ethics in AI are evolving areas, and a static understanding will quickly become outdated. In connection with our findings, this recommendation acknowledges that the weaknesses we identified are not only technical but also organizational. Addressing them requires not only changes in models and data but also growth in the knowledge, judgment, and ethical awareness of the people who design and oversee these systems.

6. Faith and Ethics Implications

The ethical implications of our project arise from the sensitive nature of toxicity detection and from the uneven model performance revealed in the analysis. A system that inaccurately identifies toxic content or disproportionately relies on identity-related features can inflict harm by allowing abusive messages to remain online or by misclassifying content that references marginalized identities. Such consequences not only affect individual users but can shape the

overall culture of online interaction. Our findings demonstrate that fairness is not automatically guaranteed in machine learning systems and that without careful monitoring, models may reproduce or amplify biases embedded within the data. These realities underscore the ethical responsibility to ensure that the systems we build treat all individuals with fairness, transparency, and respect.

The ethical benefits of engaging in this project include deepening our understanding of how bias manifests in technical systems and learning how to identify and mitigate harmful patterns. The challenges, however, highlight the difficulty of navigating fairness concerns when identity-related language is both a marker of toxicity and a potential source of unintended bias. Our project illuminated the need for careful interpretation, thoughtful feature design, and rigorous auditing. It also emphasized that high overall accuracy does not reflect the true ethical risks associated with uneven error distribution.

The Christian faith perspective further enriches this discussion by grounding our work in values of justice, impartiality, and care for the vulnerable. Scripture speaks repeatedly about the importance of fair judgment, truthfulness, and protection for those who are marginalized or mistreated. These principles align directly with the goals of ethical toxicity detection. Ensuring that harmful speech is accurately identified reflects a commitment to protecting individuals from injustice. Ensuring that identity-related content is not unfairly penalized reflects a commitment to treating all individuals with equal dignity. The work required to develop responsible AI systems mirrors the Christian call to stewardship, where individuals are entrusted with tools and knowledge that must be used to uphold the dignity and well-being of others.

At a personal level, working on this project revealed how deeply ethical considerations are woven into data analysis. The process of evaluating misclassified toxic comments, interpreting SHAP values, and questioning why identity-related terms appeared as influential predictors forced me to confront how easily harm can occur when technical systems operate without oversight. It also taught us that ethical decision-making requires humility, reflection, and a willingness to question the results. This project reminded us that fairness is not optional but integral to creating technology that aligns with moral and faith-based values.

More broadly, the ethical considerations raised by this project extend to the fields of data analytics, business, and management. In data analytics, practitioners must evaluate whether models not only perform well but also distribute errors in a way that is fair and just. In business, ethical leadership requires anticipating the impact of decisions on stakeholders and protecting individuals from unintentional harm. In management, responsible oversight involves ensuring that systems operate transparently and align with organizational values. This project illustrates that fairness cannot be an afterthought; it must be embedded in every stage of the analytical process.

This project ultimately demonstrated that ethical and faith considerations must guide the entire lifecycle of AI development, from data collection to feature design, model evaluation, and governance. It reveals the need for continuous reflection, responsible stewardship, and a commitment to justice. These values provide the foundation for developing toxicity detection systems that operate not only effectively but equitably, supporting safer online environments for all users.

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